Strict convexity of the pendulum time map

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Task 1. Resolve the time-map convexity conjecture

For $0 < \phi < \pi/2$ and $0 < z < 2$, put

$$\beta(z) = \tfrac{1}{2}\arccos(1 - z),$$

$$T_B(z, \phi) = \int_0^{\phi} \frac{dx}{\sqrt{z - \cos(2\phi) + \cos(2x)}} + \int_0^{\beta(z)} \frac{dx}{\sqrt{z - \cos(2\phi) + \cos(2x)}}.$$

Conjecture 1 of [1], retained as Conjecture 1 in the authors' postpublication manuscript of [2], asks whether this is convex in z for every fixed ϕ.

Theorem 1. *The conjecture holds. In fact, throughout the stated parameter rectangle,*

$$\boxed{\frac{\partial^2 T_B}{\partial z^2}(z, \phi) \geq \frac{\sin\phi}{16 z^{3/2}} > 0.}$$

Proof. Fix $w = 1 - \cos(2\phi) = 2\sin^2\phi \in (0, 2)$, and write $s = z + w$. In both integrals use $v = 2\sin^2 x$. Since $dx = dv/(2\sqrt{v(2 - v)})$, this gives

$$T_B(z, \phi) = \frac{1}{2}\left(\int_0^{w} + \int_0^{z}\right) \frac{dv}{\sqrt{v(2 - v)(s - v)}}. \tag{1}$$

For each integer $n \geq 0$, define

$$F_n(z, w) = \int_0^{w} \frac{v^{n - 1/2}}{\sqrt{z + w - v}}\, dv + \int_0^{z} \frac{v^{n - 1/2}}{\sqrt{z + w - v}}\, dv.$$

The binomial series and its nonnegative terms yield

$$T_B(z, \phi) = \frac{1}{2\sqrt{2}} \sum_{n=0}^{\infty} \frac{\binom{2n}{n}}{8^n} F_n(z, w). \tag{2}$$

This passage follows either by monotone convergence or by uniform convergence of the binomial series on $0 \leq v \leq \max\{z, w\} < 2$.

We prove that F_0 is constant and every F_n, $n \geq 1$, is convex as a function of $z > 0$ with $w > 0$ fixed. Indeed,

$$F_0(z, w) = 2\arcsin\sqrt{\frac{w}{s}} + 2\arcsin\sqrt{\frac{z}{s}} = \pi.$$

For $n \geq 1$, the first integral in F_n has positive second derivative

$$\frac{3}{4}\int_0^w v^{n-1/2}(z+w-v)^{-5/2}\,dv > 0.$$

Differentiation under this fixed integral is justified locally by $z+w-v \geq z > 0$. In its second integral substitute $v = zt$, obtaining

$$\int_0^1 t^{n-1/2}\frac{z^{n+1/2}}{\sqrt{w+(1-t)z}}\,dt.$$

For $b \geq 0$, direct differentiation gives

$$\begin{aligned}\frac{d^2}{dz^2}\frac{z^{n+1/2}}{\sqrt{w+bz}} &= \frac{z^{n-3/2}}{(w+bz)^{5/2}}P_n(w,bz), \qquad (3)\\ P_n(w,u) &= (n^2-\tfrac{1}{4})w^2 + (2n+1)(n-1)wu + n(n-1)u^2.\end{aligned}$$

All coefficients in this polynomial are nonnegative for integer $n \geq 1$, and its first term is positive. Thus the second integral is also convex. Its local differentiability follows from $w > 0$ and the integrable weight $t^{n-1/2}$.

Consequently the series (2) is a finite-valued pointwise limit of convex functions. Convexity passes to this limit by passing to the limit in the defining convexity inequality. To get the advertised quantitative conclusion, elementary integration gives

$$F_1(z,w) = \frac{\pi}{2}(z+w) - 2\sqrt{zw}, \qquad \partial_z^2 F_1(z,w) = \frac{\sqrt{w}}{2z^{3/2}}.$$

The function

$$T_B(z,\phi) - \frac{1}{8\sqrt{2}}F_1(z,w) = \frac{1}{2\sqrt{2}}\left(\pi + \sum_{n=2}^{\infty}\frac{\binom{2n}{n}}{8^n}F_n(z,w)\right)$$

is convex by the same reasoning. It is C^2: the original two integrals have smooth endpoints and strictly positive radicands on every compact subset of the parameter rectangle. Its second derivative is therefore nonnegative. Hence

$$\partial_z^2 T_B(z,\phi) \geq \frac{\sqrt{w}}{16\sqrt{2}\,z^{3/2}} = \frac{\sin\phi}{16z^{3/2}},$$

as required. □

Consequence. The derivative is strictly increasing. The source proves $\partial_z T_B \to -\infty$ as $z \downarrow 0$. As $z \uparrow 2$, differentiation of the original formula gives

$$2\partial_z T_B = -\left(\int_0^{\phi} + \int_0^{\beta(z)}\right)(z - \cos 2\phi + \cos 2x)^{-3/2}\,dx + \frac{1}{\sqrt{z(2-z)w}} \longrightarrow +\infty :$$

the integrals remain bounded for fixed $w > 0$. Thus there is exactly one stationary point, a nondegenerate global minimum. This establishes the single-minimum consequence for the type B branch as well.

Verification and status. The accompanying verifier checks the algebraic identity (3), the first coefficient, the changes of variables, and the series against direct quadrature. These numerical checks are supplementary; the proof above covers the entire open rectangle. The separate literature audit records the live open status, empty claim fields, current source checks, and the scope of the claim search.

Task 2. Locate the unique minimum

Determine asymptotic expansions for its location $z_*(\phi)$ and value $T_B(z_*(\phi), \phi)$ as $\phi \downarrow 0$ and $\phi \uparrow \pi/2$, with explicit error estimates.

Task 3. Resolve the branches with additional turns

Use the convex expansion together with the complete-period contribution to settle uniqueness and nondegeneracy of the minima on the higher turning branches described in [2], on their full domains.

Task 4. Extend the convexity mechanism

Characterize potential functions whose analogous change of variables produces a positive power-series weight, and establish a time-map convexity theorem for that class.

Task 5. Separate the boundary parameters

Allow the prescribed terminal velocity and initial angle to vary independently. Determine the parameter region in which the corresponding mixed-boundary time map is convex and has a unique minimum.

References

[1] F. P. da Costa, M. Grinfeld, J. T. Pinto, and K. Xayxanadasy, *Bifurcating solutions in a non-homogeneous boundary value problem for a nonlinear pendulum equation*, arXiv:1907.13009 (2019), equation (14) and Conjecture 1. `https://arxiv.org/abs/1907.13009`.

[2] F. P. da Costa, M. Grinfeld, N. J. Mottram, J. T. Pinto, and K. Xayxanadasy, *Steady state solutions in a model of a cholesteric liquid crystal sample*, Afrika Matematika **32** (2021), 645–672, doi:10.1007/s13370-020-00851-9. The authors' postpublication manuscript, including Conjecture 1, is archived at `https://strathprints.strath.ac.uk/79201/1/da_Costa_etal_AM_2021_Steady_state_solutions_in_a_model.pdf`.